

VEGA-IOT GATEWAY

Industrial IoT + AR Innovation · Embrix'26 Vegathon

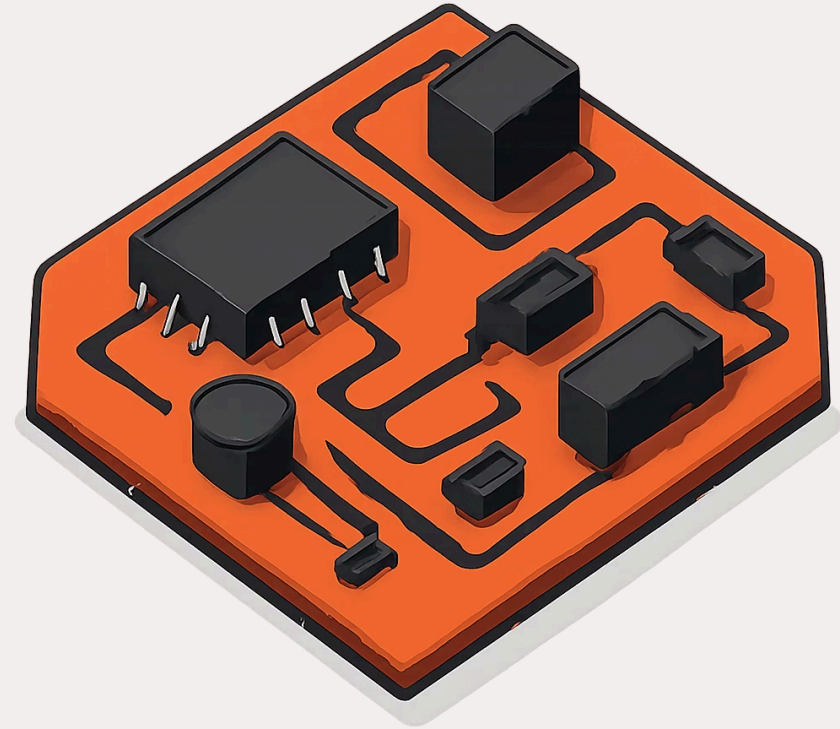
Problem

Factory floor workers lack real-time visibility into machine status — forcing manual checks and delayed decisions

Track - 2

Team

Code & Copper



THE SOLUTION



Vega Gateway

- Polls multiple machines via Modbus RTU/TCP
- Transmits over LoRa, WiFi, Bluetooth, or Ethernet
- Syncs live data to cloud dashboard



AR Smart Helmet

- Overlays real-time machine data in worker's field of view
- Worker identification and live HUD insights
- Closes the loop between data and action on the floor

📌 Most industrial IoT stops at a dashboard — Vega-IoT extends live machine data directly to the worker through AR.

TECHNICAL APPROACH

1

01 · Data Layer

Modbus via MAX485 RS-485
transceiver

2

02 · Connectivity

ESP-01 WiFi · W5500 Ethernet · LoRa ·
Bluetooth

3

03 · Cloud Backend

Aggregation, processing, APIs

4

04 · AR Helmet Stack

Edge vision on Arduino · AR sync ·
rendering

5

05 · Dashboard

Web + AR interfaces

Simultaneous polling, seamless transmission, real-time visibility.

FEASIBILITY & SCALABILITY



Feasibility

- Modbus polling on established industrial standards
- Multi-protocol transmission widely proven
- Cloud backend and dashboard on accessible stacks



Challenges & Risks

- Real-time AR sync latency
- Polling multiple machines at scale
- Toolchain maturity on indigenous hardware



Mitigation

- Multi-protocol fallback prevents single connectivity failure
- Modular architecture scales gateway and helmet independently
- Cloud-side buffering handles intermittent connectivity

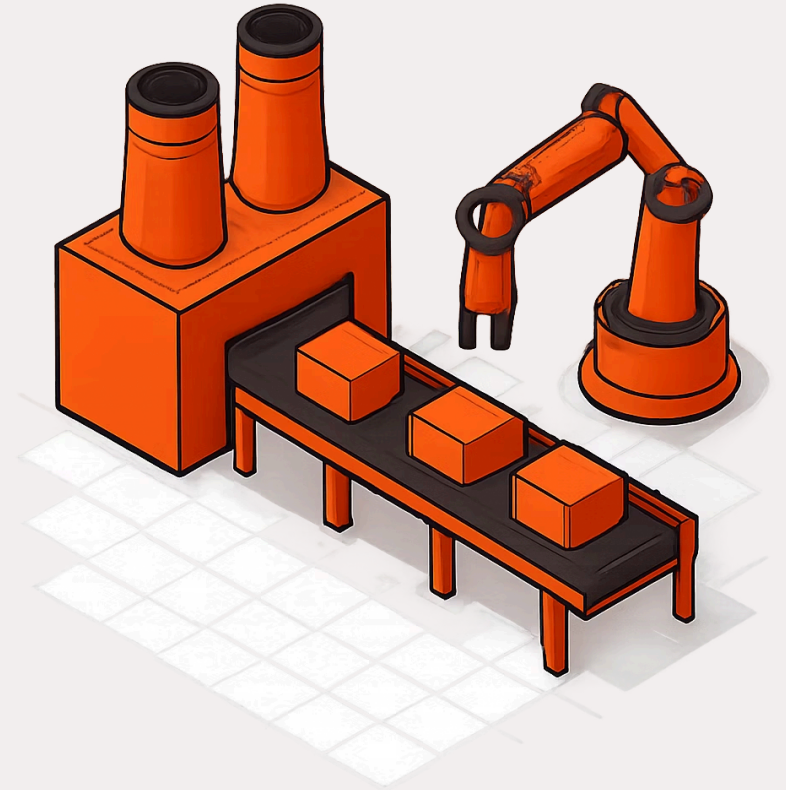
IMPACT & BENEFITS

Who Benefits

- Plant operators gain faster visibility into machine status
- Maintenance teams respond sooner to faults and downtime
- Supervisors get clearer real-time production oversight

Ecosystem Impact

- Supports LoRa, WiFi, Bluetooth, and Ethernet
- Reduces bottlenecks from multi-machine Modbus polling
- Scales across facilities of any size



RESEARCH & REFERENCES

✘ C-DAC VEGA Processor

VEGA Processor documentation —
ARIES v2.0 hardware reference

vegaprocessors.in

✘ Modbus Organization

Modbus Application Protocol
Specification — official protocol
reference

modbus.org

✘ MeitY DIR-V Programme

Digital India RISC-V (DIR-V)
Programme overview — indigenous
silicon roadmap

meity.gov.in

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